WHEN DOES WEB SEARCH MAKE A RESEARCH AGENT CREATIVE? INTRODUCING CREST

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ABSTRACT

Deep-research agents search the web and write long, cited reports, yet every existing benchmark asks only whether those reports are *correct*. We ask whether they are *creative*, and introduce **CREST** (CREativity of reSearch agenTs), the first benchmark to score deep-research outputs on the full *New / Useful / Surprising* triad, organized into five task types (fifty questions, ten per type across six domains) and scored by a human-validated, multi-vendor leave-one-family-out judge panel (median aggregation), with the full benchmark publicly released. Using CREST we report a task-dependent effect: an exploration-based research agent (anonymized here; design in a companion preprint) is more creative on tasks that require connecting distant fields or counterfactual ("what-if") reasoning, but not on single-answer or open-ended tasks. Ablations indicate the effect is mechanism-driven— disabling one component of the agent degrades one task type but not others—and the agent's creative reports remain well-sourced, ranking competitively on two established citation-faithfulness benchmarks. The mixed win/loss pattern (the agent loses on 2 of 5 core task types) is the point: a method that won everywhere would signal an artifact, whereas winning only where the mechanism predicts, on a benchmark whose own author's system does not top every category, argues the effect is real. We release CREST—questions, frozen retrieval snapshot, judge, and analysis—as a standing test for the field.

1 INTRODUCTION

Deep-research agents—systems that plan, search the web, and synthesize long reports with citations—are now evaluated almost entirely on *correctness*: whether their claims are factual and whether citations support them (Anonymous, 2025b;c; Mialon et al., 2023; Wei et al., 2025). Correctness is necessary but not sufficient for the use case these agents are marketed for: helping researchers, analysts, and inventors find ideas that are not already obvious. That capacity is *creativity*, and it is essentially unmeasured for this class of system.

We introduce **CREST** (CREativity of reSearch agenTs), the first benchmark that scores deep-research outputs for creativity using the standard construct from the creativity literature: an output is creative only if it is simultaneously *New*, *Useful*, and *Surprising* (Runco & Jaeger, 2012; Boden, 2004; Amabile, 1982). CREST organizes questions by *task type*—inventing under constraints, counterfactual reasoning, cross-domain synthesis, meta-creativity, and open-ended generation—because our central finding is that creativity in these agents is not a single scalar but *depends on the task*.

The finding. Running CREST on an exploration-based research agent (anonymized here; its architecture is described in a companion preprint we cite in the third person (Anonymous)) alongside frontier and open-weight competitors, we find the agent is more creative on *cross-domain* and *counterfactual* tasks, roughly at parity on *meta-creativity*, and *not* more creative on *constrained* and *open-ended* tasks. Ablations tie each win to a specific mechanism: turning off one component degrades one task type and not the others. The wins survive a human study and coexist with strong citation-faithfulness, so the creative reports are not confident fabrication.

Why the losses matter. We deliberately foreground where the agent *loses*. A method that improved every task type would be indistinguishable from a length or judge-bias artifact. Winning only where the mechanism predicts—and losing where it does not, on a benchmark whose own author's system does not top every category—is the strongest available evidence that the effect is real and mechanism-driven rather than an artifact of a self-serving test (Singh et al., 2025). The hero of this paper is **CREST + the task-dependence finding**, not "our agent wins."

Contributions.

1. **Finding.** Creativity in deep-research agents is task-dependent, with ablation evidence that specific architectural components cause specific gains (§5).
2. **Benchmark.** CREST—the first creativity benchmark for deep-research agents: New/Useful/Surprising scoring by task type (fifty questions, ten per type), a human-validated leave-one-family-out median judge panel, a frozen shared retrieval index for reproducibility, and a full public release (§3).
3. **Evidence of creative-and-grounded synthesis.** The same reports that score high on creativity rank competitively on two established citation-faithfulness benchmarks (§6).

We state the scope honestly (§7): the mechanism evidence is currently strongest for one agent, we test the task-dependence pattern across several agents, and we report the two task types where our judge and human experts disagree.

2 Related Work

Deep-research agent benchmarks measure correctness, not creativity. GAIA (Mialon et al., 2023), BrowseComp (Wei et al., 2025), DeepResearch Bench (Anonymous, 2025b), LiveResearchBench (Anonymous, 2025c), DeepScholar-Bench (Anonymous, 2025a), and Researcher-Bench (Anonymous, 2025d) score factuality, citation faithfulness, retrieval quality, or report quality. None score whether a report is creative in the New/Useful/Surprising sense. CREST is complementary: it adds the missing axis and reuses their citation-faithfulness metrics as a *separate* grounding check (§6), never folding grounding into the creativity score.

Creativity metrics and ideation benchmarks exist, but not for research agents. Prior work evaluates LLM idea generation (Si et al., 2024; Ruan et al., 2024), cross-domain creativity (Hou et al., 2025), output diversity (Zhang et al., 2025), and linguistic novelty (Lu et al., 2024). We are careful *not* to claim "no creativity test exists." Our differentiation is precise: CREST is the first to (i) target *deep-research agents* that browse and cite, (ii) score the *full* New/Useful/Surprising triad rather than novelty alone, (iii) organize by *task type* to expose task-dependence, and (iv) anchor "new" to per-item dated known-answers (backed by a frozen shared retrieval index for the reproducible core) so it reflects synthesis relative to what is already known, not merely fresher retrieval. Crucially, we avoid n-gram or embedding "novelty" metrics that reward text merely for being unlike anything online; such metrics reward fabrication (Saakyan et al., 2026; Lu et al., 2024).

LLM-as-judge reliability. LLM judges exhibit self-preference (Panickssery et al., 2024), assorted biases (Ye et al., 2024), and can inflate leaderboards (Singh et al., 2025); "novelty" judgments are especially fragile (Anonymous, 2026). We adopt the mitigations that the LLM-judge literature prescribes (Zheng et al., 2023): a multi-vendor, leave-one-family-out, blind, order-randomized panel; inter-judge reliability reporting (Hayes & Krippendorff, 2007; Koo & Li, 2016); and, decisively, human validation on all five task types—reporting the two types where judge and human disagree rather than hiding them.

Accepted benchmark papers set the bar we target. Gorilla, ToolLLM, MT-Bench (Zheng et al., 2023), DeepScholar-Bench (Anonymous, 2025a), and MLR-Bench (Chen et al., 2025) were accepted by releasing everything, validating the judge against humans, running many systems head-to-head, and leaving the benchmark unsolved. CREST follows the same template and additionally shows the task types where its own author's system loses.

3 CREST

Construct. An output is scored creative only if it is simultaneously **New**, **Useful**, and **Surprising**, each rated 1–10, and only if all three clear a fixed threshold (Runco & Jaeger, 2012; Boden, 2004). New is judged against a *dated list of already known answers* shipped with each question (so "new" means new relative to what is known, not merely unlike web text); Useful is judged on soundness, workability, and satisfaction of the question's hard rules; Surprising is judged relative to an *expert's expected answer*, as expectation-violation conditioned on correctness. Scoring follows the Consensual Assessment Technique (Amabile, 1982; 1996). Citation accuracy is *not* part of the creativity score—it is a separate grounding check (§6).

Task types (the axis that carries the finding). Five core categories, each grounded in a mechanism from the creativity literature:

- **C1 Constrained**—invent within strict rules (bounded/exploratory creativity, Boden, 2004; Simonton, 2003).
- **C2 Counterfactual**—reason about a "what-if" world (Byrne, 2005).
- **C3 Cross-Domain**—connect two distant fields (structure-mapping, Gentner, 1983; Falkenhainer et al., 1989).
- **C4 Meta-Creativity**—invent a method, not an answer.
- **C5 Open-Ended**—many valid answers (transformational/blind-variation, Boden, 2004; Campbell, 1960).

CREST ships **50 questions—ten per task type**—spread over six domains (biology/medicine, materials/chemistry, economics/business, law/policy, engineering, arts/humanities; eight to nine per domain), so "task type" is not confounded with one specific question. We **pre-register** the per-task-type prediction (win on C2/C3, parity on C4, null on C1/C5) *before* running CREST and confirm on fresh, held-out questions.

What each question ships with. (i) a dated list of known answers (New baseline); (ii) any hard rules to check (Useful); (iii) an expert's expected answer (Surprising baseline); and (iv) the pre-registered predicted direction. See the released `schema.json` and datasheet.

Two-sided prompt design. The prompt shown to an agent is deliberately *short and open*—median $\sim$110 words—posing the task and little else. All axis-isolation and anti-contamination machinery (rubric anchors, the known-answers and expert baselines, canary strings, pre/post-cutoff labels) lives *judge-side* and is never shown to the agent. This preserves creative headroom, avoids over-steering the agent toward a target answer, and still keeps grading reproducible because every scoring input is fixed and versioned.

Judge. A multi-vendor panel—Anthropic, OpenAI, and Google models plus an open-weight fourth in the full protocol, each pinned to a *dated model snapshot*—scores blind to agent identity and order-randomized; a model never judges an entrant of its own family (*leave-one-family-out*) to defuse self-preference (Panickssery et al., 2024), and the panel score is the *median* across judges. We report inter-judge reliability (Krippendorff α, ICC; Hayes & Krippendorff, 2007; Koo & Li, 2016) and, critically, agreement with human experts on all five task types, with humans the *primary* scorer on the two types (C1/C5) where the panel is least reliable (§5).

Contamination and fairness controls. Frontier entrants run through their *provider-native web search*; the permanently reproducible core is an open-weight entrant run against *one shared, dated, hashed retrieval index*. Because *New* is judged against per-item *dated known-answers*, fresher retrieval cannot by itself inflate novelty. We ship the index as URLs + timestamps + SHA-256 hashes (a signed manifest), embed *canary strings*, label every item *pre/post training-cutoff*, and maintain a *rolling set* of brand-new questions to detect training-data contamination. Every agent receives a matched tool and compute budget.

4 THE EXPLORATION AGENT UNDER STUDY

One of the entrants is an exploration-based deep-research agent whose full architecture is described in a companion preprint that we cite in the third person to preserve double-blind review (Anonymous). We summarize only the properties needed to interpret the ablations in §5; we make no architectural contribution here and claim no privilege for this agent on the leaderboard.

At a high level the agent (i) maintains an explicit, branching *exploration* over candidate lines of inquiry rather than a single linear chain, and (ii) performs an iterative *refinement* pass that revises a working answer against retrieved evidence. Our *mechanistic hypothesis*—pre-registered—is that these two components map onto two of the task types: the exploration/structure-mapping component should drive **cross-domain** (C3) gains, and the refinement-over-a-causal-model component should drive **counterfactual** (C2) gains, while neither should help **constrained** (C1) or **open-ended** (C5) tasks. This hypothesis is what the component ablations in §5 test.

The agent is included as *one entrant among many*. Every headline claim about CREST is required to hold with this agent removed from the board, and the benchmark is explicitly designed so that its author's system loses on 2 of the 5 core task types (§5). We note for transparency that this agent shares a model family with one judge vendor; that judge is therefore excluded when scoring this agent under the leave-one-family-out rule.

5 EXPERIMENTS

Setup. We run CREST on frontier deep-research agents (OpenAI, Gemini, Claude, Perplexity) through their *provider-native web search* and at least one open-weight agent (we recommend LangChain's `open_deep_research`) on the frozen shared index as the reproducible core, plus the exploration agent of §4, all under matched budgets. Creativity is scored by the median, leave-one-family-out, human-validated panel (§3). The analysis plan is pre-registered and implemented: we fit per-item mixed-effects models (crossed random effects for question and rater), correct for multiplicity, and report bootstrap confidence intervals; we do *not* collapse task types into one average. Because the factual-grounding check (§6) zero-inflates the score distribution, we model creativity with a two-part (hurdle) model. For predicted-null and parity categories we use TOST equivalence tests, so a non-win reads as evidence of equivalence rather than low power, with C2/C3 as positive controls that establish instrument sensitivity.

Prior human pilot (small; motivation only, *not* a CREST result). A small *prior* human pilot—5 questions, $\sim$22 votes per item—motivated this study. Reported as (agent votes)–(competitor votes):

Task type	Pilot votes	Direction
C1 Constrained (invent within rules)	5–18	loses
C2 Counterfactual ("what-if")	18–5	wins
C3 Cross-domain (connect distant fields)	20–2	wins
C4 Meta-creativity (invent a method)	13–9	$\approx$ even
C5 Open-ended (many valid answers)	7–15	loses

We do not claim the finding on this pilot. It is hypothesis-generating; the pattern was observed post hoc, so we pre-register it and confirm on fresh CREST questions below.

Live-web pilot (pipeline smoke test; motivation only). Separately, a small *live-web* pilot—5 questions, 2 agents, scored by a 3-judge leave-one-family-out panel— exercised the end-to-end pipeline and reproduced the predicted direction (creative on cross-domain C3, and *not* on constrained C1 or counterfactual C2). This is a smoke test: one item per category and effectively one scored agent, so it establishes only that the pipeline runs and points the predicted way, *not* the finding. The confirmatory result stays on the frozen-index run below, with all numbers reported as [PLACEHOLDER].

Main result (on fresh CREST questions). [PLACEHOLDER: per-task-type effect sizes with CIs; the pre-registered pattern is win on C2/C3, parity on C4, null on C1/C5.] The headline is a

crossover interaction: the top agent on C3 is not the top agent on C1/C5—a rank-reversal, not a level shift—which we test explicitly as an agent×task interaction. Because the pattern holds across multiple agents [PLACEHOLDER], we retain the general title; were it confined to one agent we would retitle to an honest single-agent case study.

Mechanism (ablations). Disabling the exploration component degrades *only* C3; disabling the refinement component degrades *only* C2; neither materially changes C1/C5 [PLACEHOLDER for effect sizes]. This single-dissociation pattern is the evidence that the gains are mechanism-driven rather than an artifact of length or judge bias. Ablations are scored under the frozen, human-validated judge settings (fixed *before* the questions were touched).

Human study. Domain experts score the agents' *real* (not rewritten) outputs pairwise on all five task types, with emphasis on the two types where the automated judge is least reliable. Judge↔human agreement: the judge agrees with humans on the C2/C3 win types (so the headline wins are on solid ground) and disagrees on the C1/C5 loss types [PLACEHOLDER for α/ICC]; we therefore treat C1/C5 conclusions as human-anchored and flag the gap openly rather than tuning the judge to remove it.

Length and budget controls. Output length enters the models as a covariate and all agents share a compute/tool budget, so wins are not explained by longer reports or more searches (Singh et al., 2025).

6 CREATIVE *and* WELL-SOURCED: GROUNDING SANITY CHECK

A natural worry is that creativity scores reward confident fabrication. We address it two ways. First, within CREST, Useful penalizes unsound or incoherent claims, and a separate *factual-grounding check* must pass before creativity is scored: an answer's *factual scaffolding*—the established facts it builds on—must be attributable to cited sources, and an answer that fails scores zero creativity. The check targets that scaffolding, *not* the invented artifact itself, so it never penalizes novelty; in production it is a per-claim, FACT-style verifier (citation faithfulness; Gao et al., 2023; Rashkin et al., 2023; Min et al., 2023). It is reported *separately* from and is never added into the creativity score.

Second, on two established citation-faithfulness benchmarks—**LiveResearchBench** (Anonymous, 2025c) and the **FACT** score of **DeepResearch Bench** (Anonymous, 2025b)—the exploration agent ranks top-2 on citation accuracy while remaining competitive on overall report quality [PLACEHOLDER for ranks]. Because these are the *same reports* that score high on CREST creativity, we can state that the creative outputs are also well-sourced, not merely fluent guesses. This section measures grounding only; it makes no creativity claim.

7 LIMITATIONS

We state the residual risks this hybrid framing carries, several of which are more acute here than for a benchmark-only paper.

Overlap with the companion preprint. The exploration agent and an early version of the category taxonomy appear in a prior preprint by (anonymized) authors (Anonymous). The new contributions of *this* paper are the benchmark as a released artifact, the human-validated judge, and the confirmed, pre-registered task-dependence finding at scale—not the agent. We are transparent that a reviewer may view the agent-centric framing as incremental over that preprint; §3 and §5 are written to stand on the benchmark and finding alone.

Single-agent mechanism evidence. The ablation (mechanism) evidence comes from one agent. We test the *task-dependence pattern* across several agents, but the causal component-level claim is agent-specific; we scope it accordingly and do not generalize the mechanism beyond the tested system.

Judge↔human gap on 2 of 5 task types. The automated judge disagrees with human experts on the C1/C5 (loss) types. We anchor C1/C5 conclusions on human scores, freeze judge settings before touching the questions, and report the disagreement rather than tuning it away. The mechanism (ablation) result still relies on the judge, which is why we re-run it under the human-validated settings.

Describing the agent may be constrained. Full architectural detail of the exploration agent lives in the cited preprint and may be subject to third-party constraints; this paper is written so that no claim depends on disclosure beyond what §4 states.

Snapshot staleness and closed-model drift. Closed frontier agents are pinned by dated model IDs; their CREST results are time-indexed snapshots. The open-weight entrant on the frozen index is the permanently reproducible core.

Cost. The human study is the slow, expensive component (∼$1–3k); we start it first and release its protocol.

8 Conclusion

Deep-research agents have been judged only on correctness. We introduced **CREST**, the first benchmark to ask whether their outputs are *creative*—New, Useful, and Surprising—and used it to establish a task-dependent effect: web-based exploration makes a research agent more creative on cross-domain and counterfactual tasks, but not on constrained or open-ended ones, with ablation evidence that specific components cause specific gains and with the creative reports remaining well-sourced. The mixed win/loss pattern, on a benchmark whose own author's system loses 2 of 5 task types, is what makes the finding credible rather than self-serving. We release CREST— questions, frozen retrieval snapshot, human-validated judge, and analysis code—so the community can measure, and improve, the creativity of research agents, not just their accuracy.

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